

Artificial Intelligence and Machine Learning Lab

Enrolment No: _____

Division: _____

Roll No: _____

Name: _____

Experiment No.: 07

Title: Implement Linear regression on any dataset. Write a program to demonstrate the working of Linear Regression .Use an appropriate data set and visualize the results.

Theory:

Simple Linear regression :

Machine Learning is a branch of Artificial intelligence that focuses on the development of algorithms and statistical models that can learn from and make predictions on data. Linear regression is also a type of machine-learning algorithm more specifically a supervised machine-learning algorithm that learns from the labelled datasets and maps the data points to the most optimized linear functions, which can be used for prediction on new datasets.

What is Linear Regression?

Linear regression is a type of supervised machine learning algorithm that computes the linear relationship between the dependent variable and one or more independent features by fitting a linear equation to observed data.

When there is only one independent feature, it is known as Simple Linear Regression, and when there are more than one feature, it is known as Multiple Linear Regression.

Types of Linear Regression

There are two main types of linear regression:

Simple Linear Regression :

This is the simplest form of linear regression, and it involves only one independent variable and one dependent variable. The equation for simple linear regression is:

$$Y = c + mX$$

where:

Y is the dependent variable

X is the independent variable

c is the intercept

m is the slope



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Multiple Linear Regression :

This involves more than one independent variable and one dependent variable.
The equation for multiple linear regression is:

$$Y = c + m_1X_1 + m_2X_2 + \dots$$

where:

Y is the dependent variable

$X_1, X_2, \dots, X_n$ are the independent variables

c is the intercept

$m_1, m_2, \dots, m_n$ are the slopes

What is the best Fit Line?

Our primary objective while using linear regression is to locate the best-fit line, which implies that the error between the predicted and actual values should be kept to a minimum. There will be the least error in the best-fit line.

The best Fit Line equation provides a straight line that represents the relationship between the dependent and independent variables. The slope of the line indicates how much the dependent variable changes for a unit change in the independent variable(s).

Key Metrics:

- Mean Squared Error (MSE): Measures prediction error.
- R-squared (R): Indicates the proportion of variance in Y explained by X.

Procedure

1. Data Preparation:

- Import or create a dataset with one independent and one dependent variable.
- Perform data cleaning and visualization.

2. Model Implementation:

- Calculate the slope (m) and intercept (c) using the formulas or regression algorithms.
- Train a simple linear regression model on the dataset.

3. Prediction:

- Use the regression model to predict values of Y for given X

4. Evaluation:

- Calculate evaluation metrics such as MSE/R squared.
- Plot the regression line over the data points.

	5. Visualization: ○ Visualize the relationship between X and Y, including the regression line.
Pseudo code or Program	<pre> import pandas as pd from sklearn.model_selection import train_test_split from sklearn.linear_model import LinearRegression import matplotlib.pyplot as plt import seaborn as sns from sklearn.metrics import mean_squared_error, r2_score # Load the dataset file_path = "coffee_shop_revenue.csv" df = pd.read_csv(file_path) # Display basic information and first few rows df.info(), df.head() # Select features and target variable X = df[["Number_of_Customers_Per_Day"]] y = df["Daily_Revenue"] # Split into training and testing sets (80% train, 20% test) X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) # Train the Simple Linear Regression model model = LinearRegression() model.fit(X_train, y_train) # Predict revenue on test data y_pred = model.predict(X_test) # Plot the regression line plt.figure(figsize=(10, 6)) sns.scatterplot(x=X_test["Number_of_Customers_Per_Day"], y=y_test, label="Actual Revenue", color="blue", alpha=0.6) sns.lineplot(x=X_test["Number_of_Customers_Per_Day"], y=y_pred, color="red", label="Regression Line") plt.xlabel("Number of Customers Per Day") plt.ylabel("Daily Revenue") plt.title("Simple Linear Regression: Customers vs Revenue") plt.legend() plt.show() # Calculate Mean Squared Error (MSE) mse = mean_squared_error(y_test, y_pred) </pre>

```
# Calculate R-squared (R²) score
r2 = r2_score(y_test, y_pred)

print(mse, r2)

#multiple linear regression
feature_cols = columns[:-1]
target_col = columns[-1]

X = df[feature_cols]
y = df[target_col]

train_X, validate_X, train_y, validate_y = train_test_split(X, y, test_size =
0.2, random_state = 1)
model = LinearRegression()
model.fit(train_X, train_y)

predicted_sales = model.predict(validate_X)

mae = mean_absolute_error(predicted_sales, validate_y)
print(mae)
```

Outputs

(Students can attach extra pages if necessary)



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(Students can attach extra pages if necessary)

Exercise for Practice

Q1.	Write the general form of the linear regression equation and explain its components ?
Ans:	
Q2.	What does a high R Square value indicate in a regression model?
Ans	
Q3	How do you calculate the coefficients m and c for simple linear regression?
Ans	
Q4.	In what scenarios is simple linear regression most suitable?
Ans	



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Conclusion	Thus, we have successfully Studied Linear Regression and visualized it.
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Date of Submission:	
Marks out of 10	Name and Sign of Subject Teacher
Remark:	